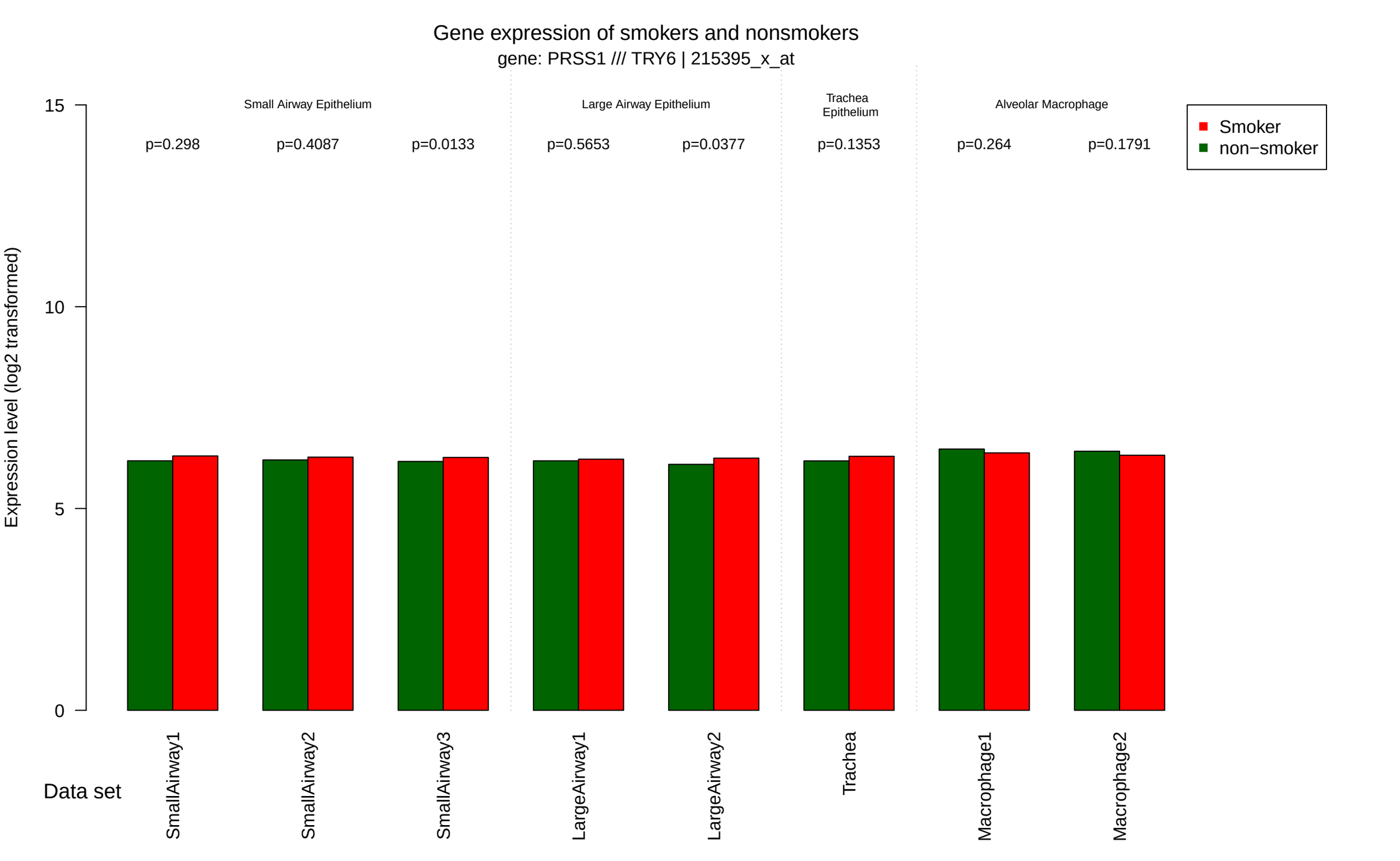
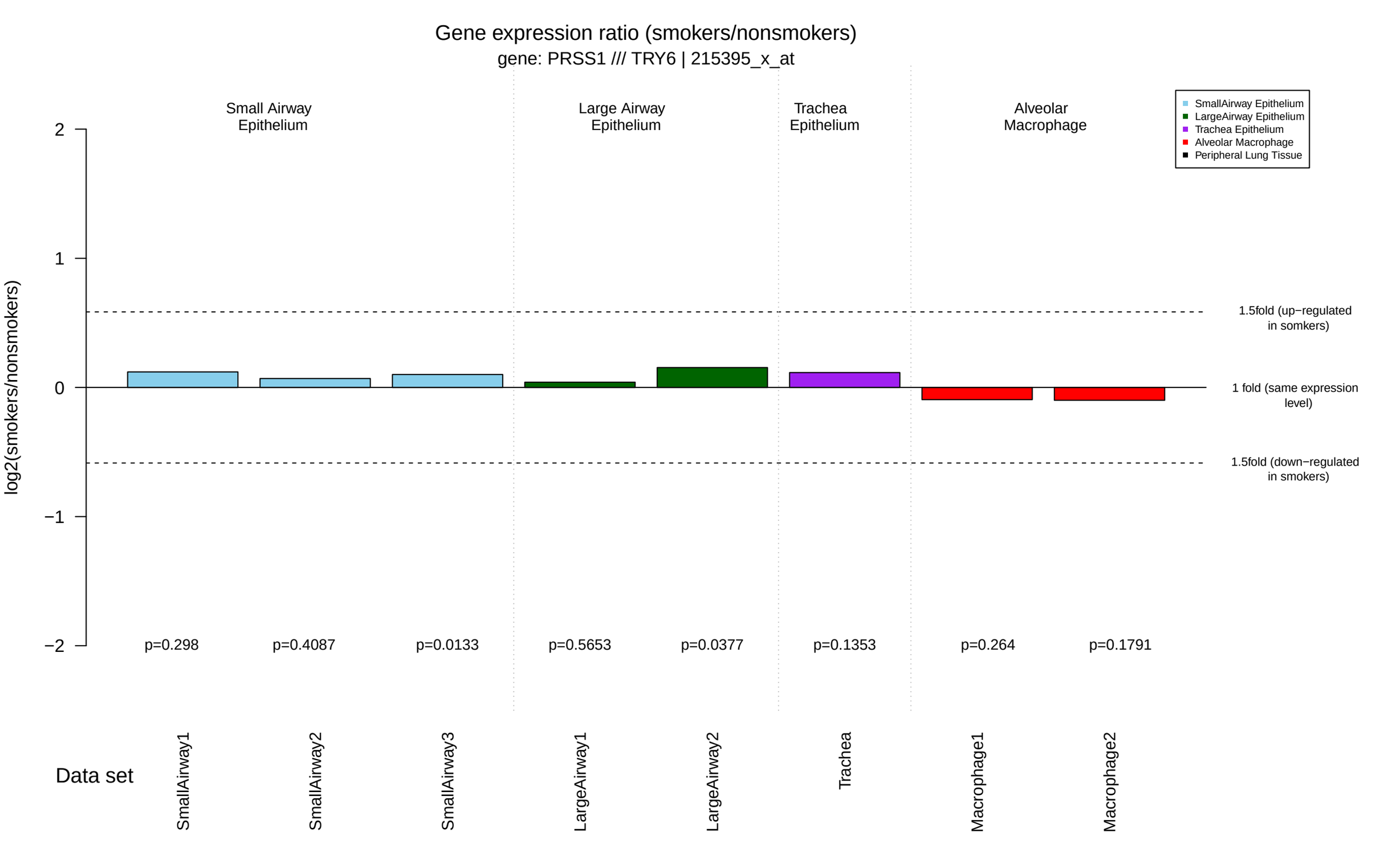
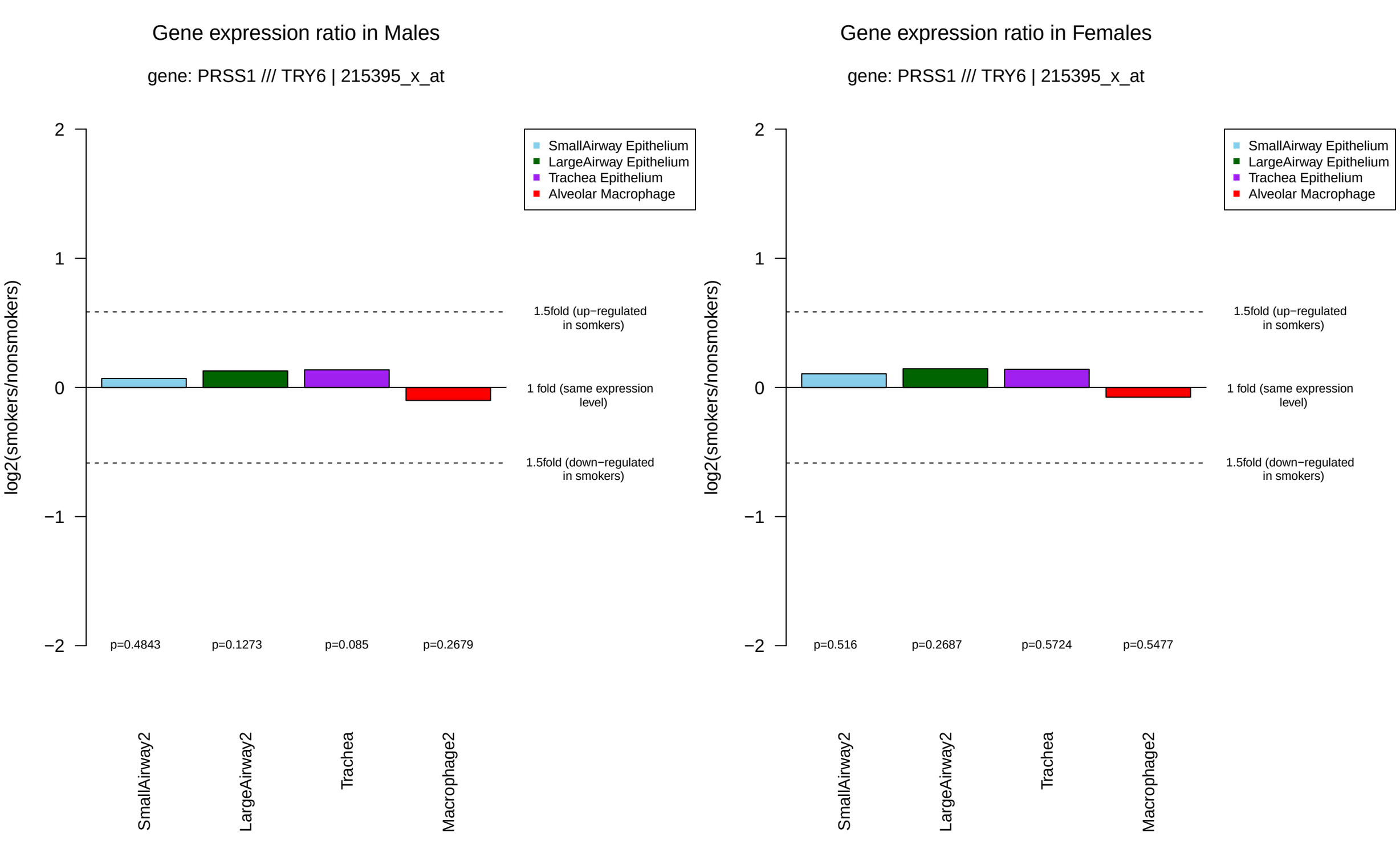
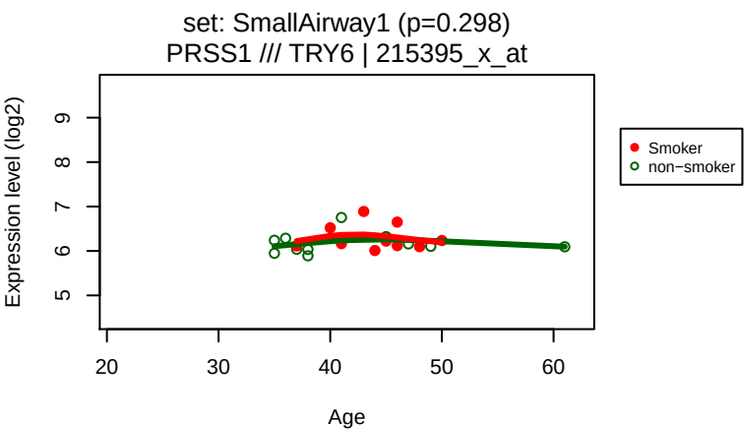
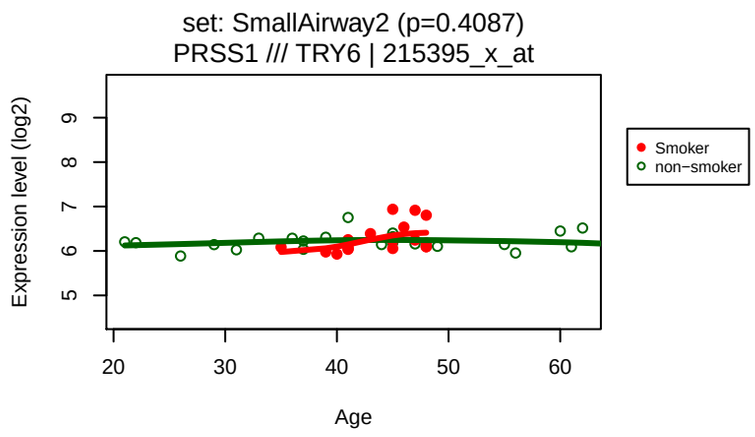
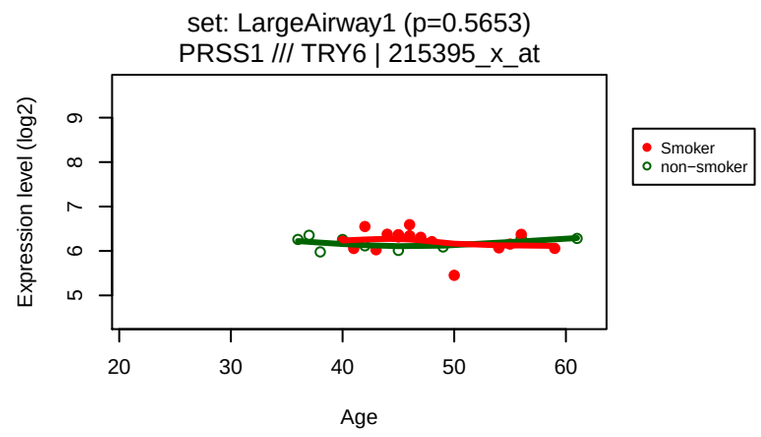

Gene expression of smokers and nonsmokers
gene: PRSS1 /// TRY6 | 215395_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.298
p=0.4087
p=0.0133
p=0.5653
p=0.0377
p=0.1353
p=0.264
p=0.1791
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

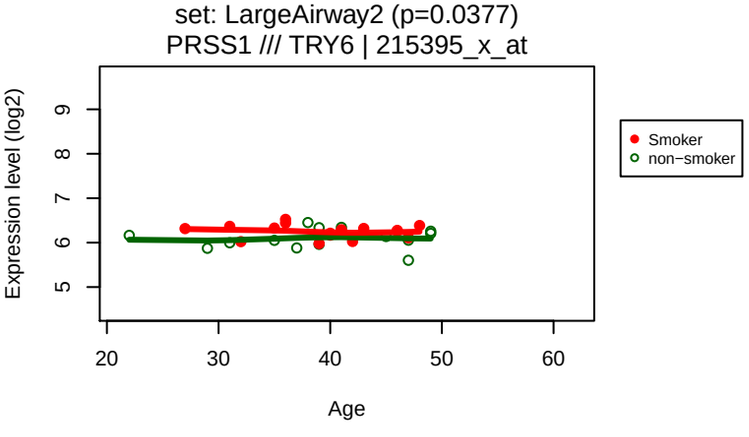

Gene expression ratio (smokers/nonsmokers)
gene: PRSS1 /// TRY6 | 215395_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.298
p=0.4087
p=0.0133
p=0.5653
p=0.0377
p=0.1353
p=0.264
p=0.1791
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

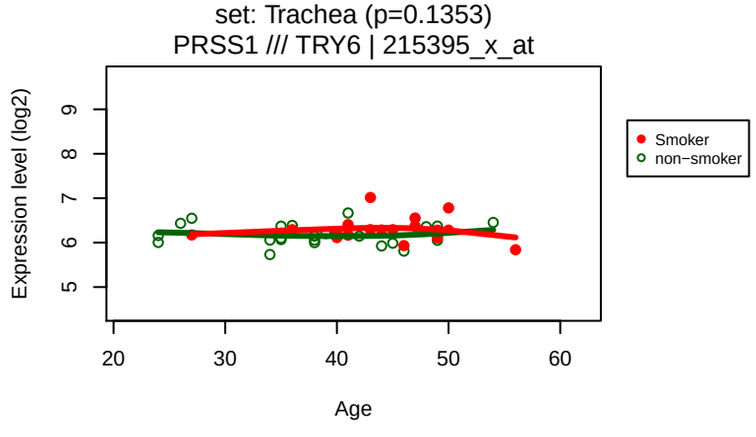

Gene expression ratio in Males
gene: PRSS1 /// TRY6 | 215395_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4843
p=0.1273
p=0.085
p=0.2679
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS1 /// TRY6 | 215395_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.516
p=0.2687
p=0.5724
p=0.5477
SmallAirway2
LargeAirway2
Trachea
Macrophage2

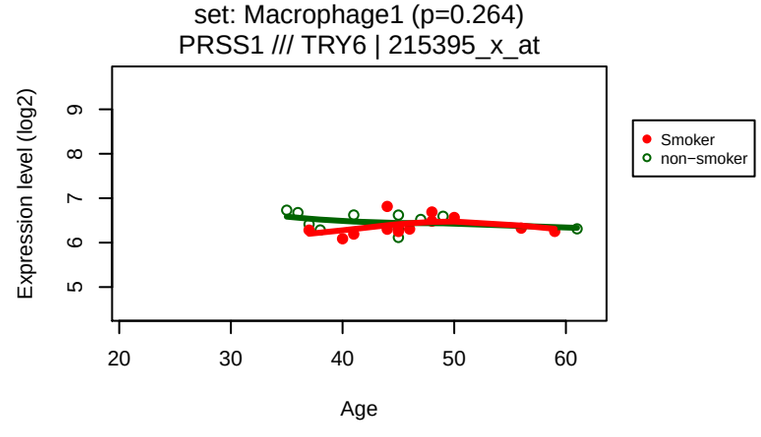

set: SmallAirway1 (p=0.298)
PRSS1 /// TRY6 | 215395_x_at
Expression level (log2)
Age
Smoker
non−smoker

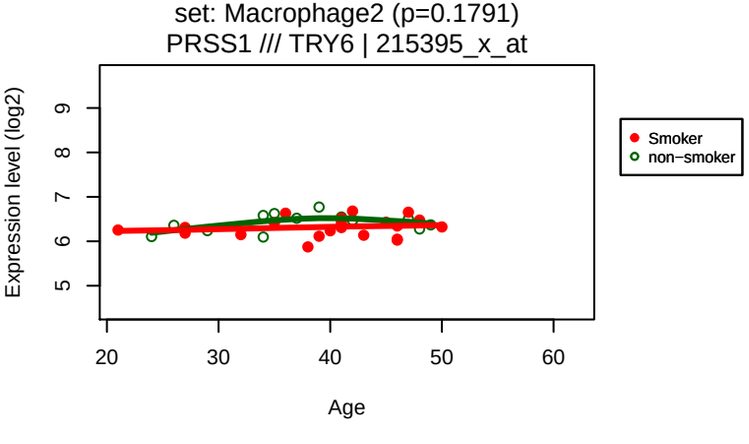

set: SmallAirway2 (p=0.4087)
PRSS1 /// TRY6 | 215395_x_at
Expression level (log2)
Age
Smoker
non−smoker

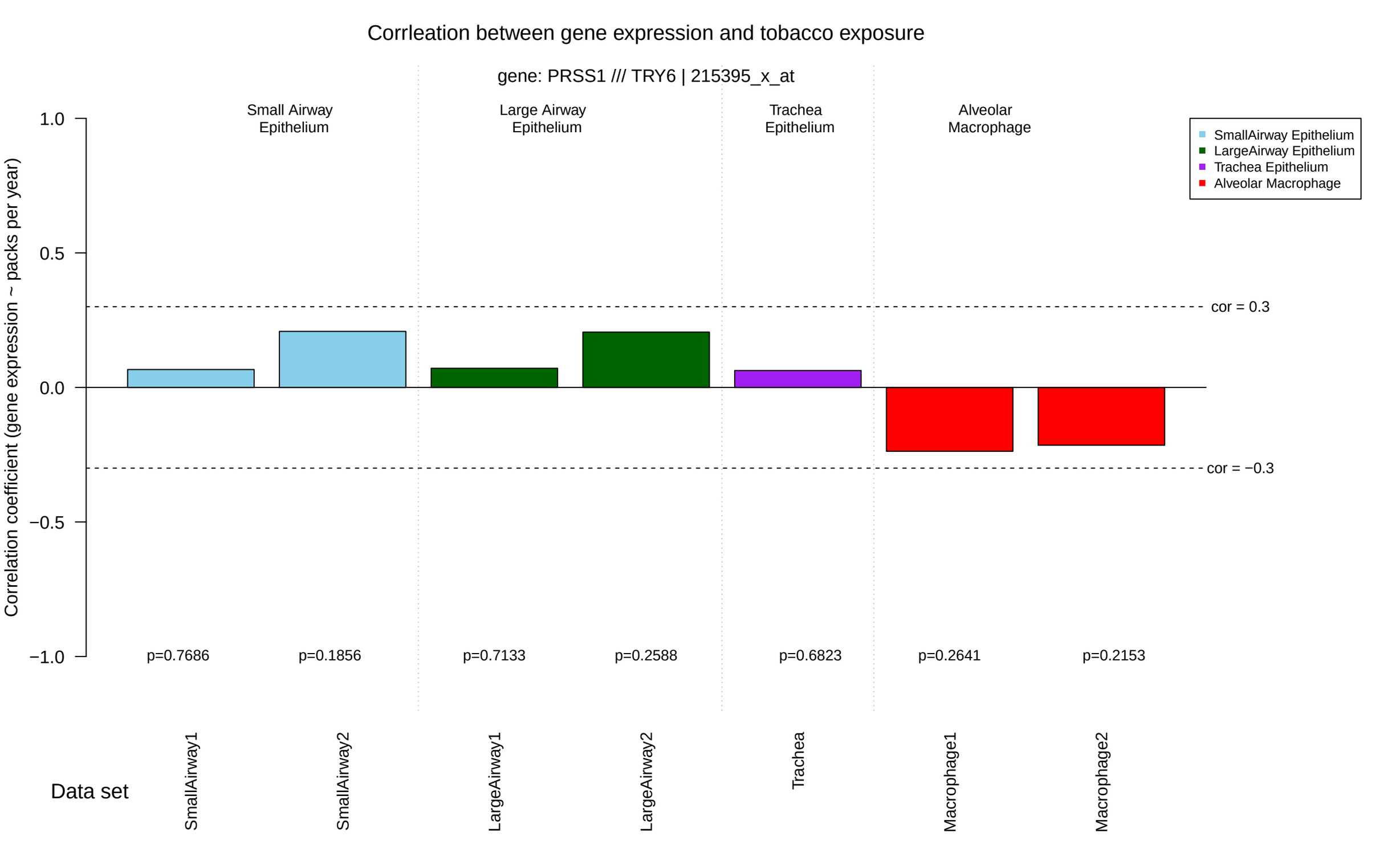

set: LargeAirway1 (p=0.5653)
PRSS1 /// TRY6 | 215395_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0377)
PRSS1 /// TRY6 | 215395_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1353)
PRSS1 /// TRY6 | 215395_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.264)
PRSS1 /// TRY6 | 215395_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1791)
PRSS1 /// TRY6 | 215395_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS1 /// TRY6 | 215395_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7686
p=0.1856
p=0.7133
p=0.2588
p=0.6823
p=0.2641
p=0.2153
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2